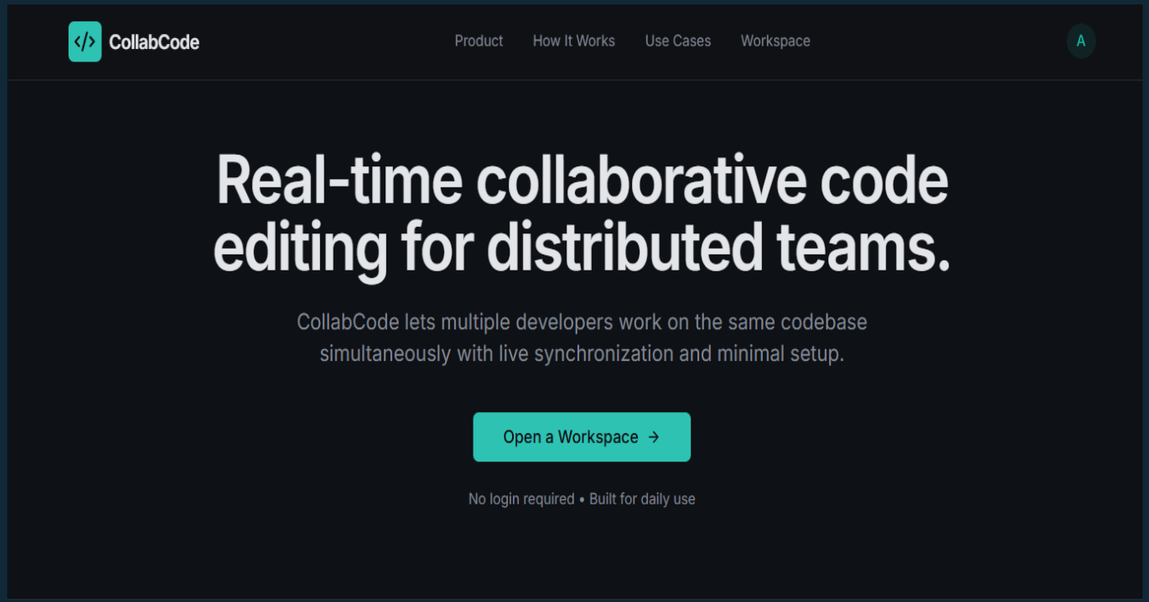
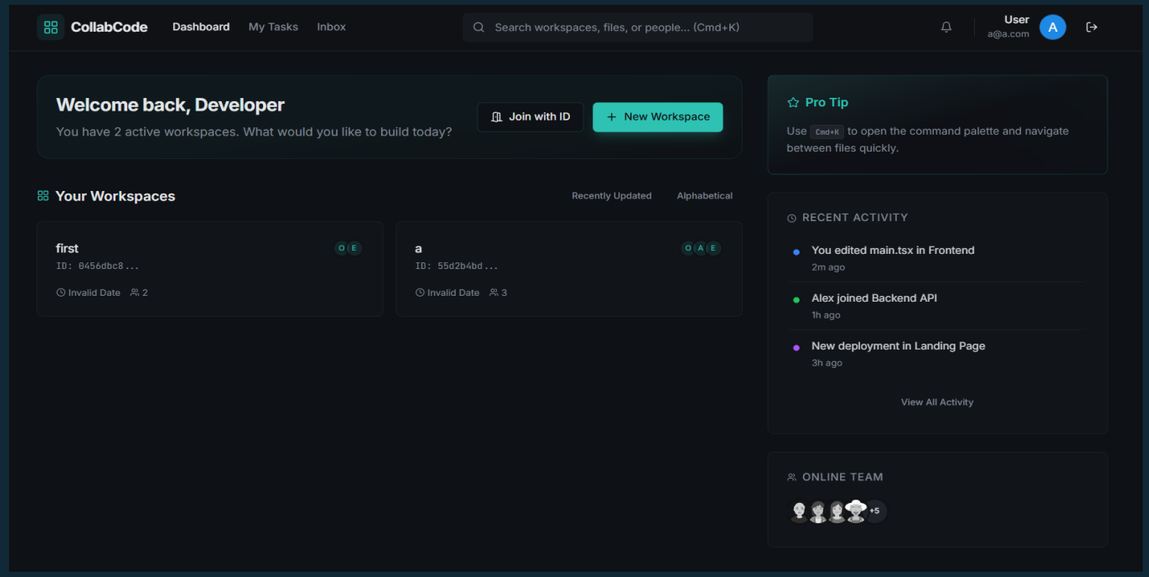
CollabCode: Elevate Your Team's Coding Workflow



The Persistent, Async-First Code Collaboration Platform

- Collaborate instantly in a powerful, browser-based workspace.
- Seamlessly work together in real-time or asynchronously.
- Keep your team connected and productive, even when offline.



- Team Name: Code_X

⚠ PROBLEM

Modern Dev Collaboration: Broken by Design

Collaboration should not depend on everyone being online at the same time.

Today's tools fail developers by:

- **Environment Drift:** Complex local setups and inconsistent configurations across machines waste hours.
- **Asynchronous Blocker:** Collaboration tools demand everyone stays online, crippling async workflows.
- **Fragile Live Sessions:** A single disconnect from the host ends the session, locking away work.

Critical Issue: When the host disconnects, the session ends and work becomes completely inaccessible.

Why Existing Solutions Are Not Enough

VS Code Live Share

Real-time only; progress is lost if the host disconnects.

GitHub

Built for version control, not for synchronous live editing.

Google Docs

Optimized for documents, inadequate for code-specific workflows.

Lightweight Web Editors

Prone to data loss and lacks robust recovery features.

> There is no tool focused on **persistent, async-first coding collaboration**.

SOLUTION

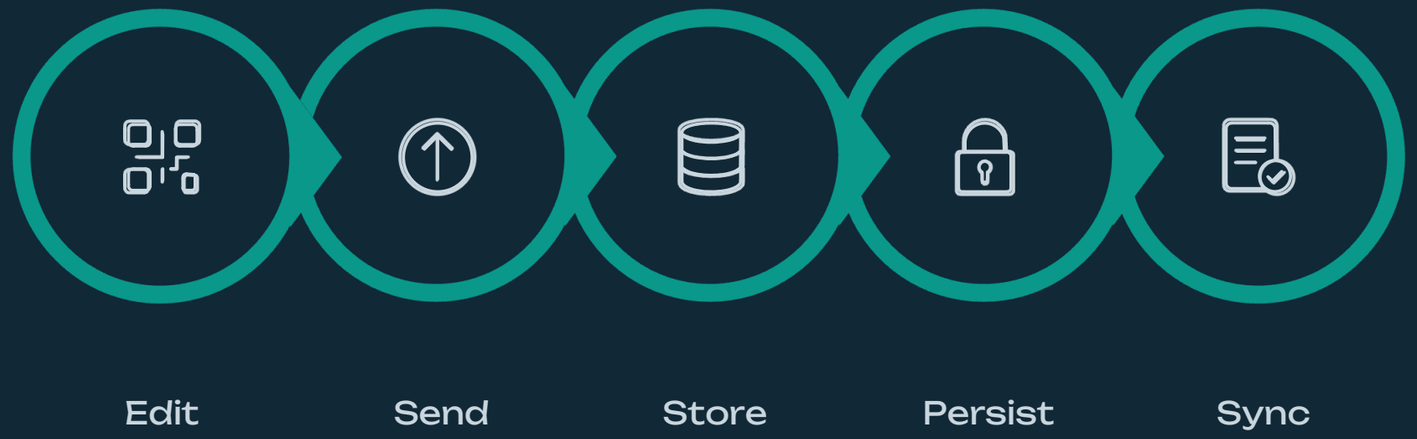
CollabCode: Revolutionizing Team Coding

A centralized, browser-based coding workspace designed for **persistent, async-first collaboration** and unparalleled team efficiency.

- **Seamless Real-time Collaboration**
  Work together effortlessly, just like you're in the same room.
- **Uninterrupted Asynchronous Productivity**
  Continue coding and contributing even when teammates are offline.
- **Always Synchronized Codebase**
  Ensures everyone always accesses and works on the latest saved version.
- **Secure Cloud Workspace**
  All your code is safely stored and managed on a robust backend.

CollabCode combines the intuitive editing experience of VS Code with the robust persistence and accessibility of a cloud platform.

How CollabCode Works



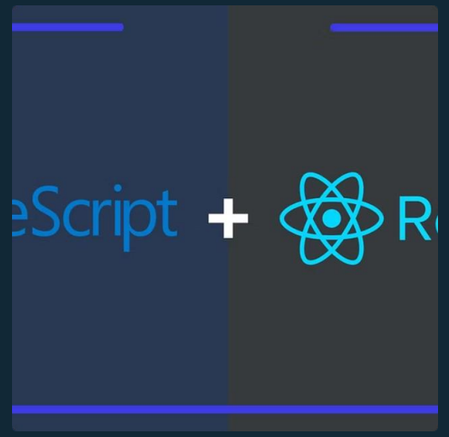
Seamless Workflow

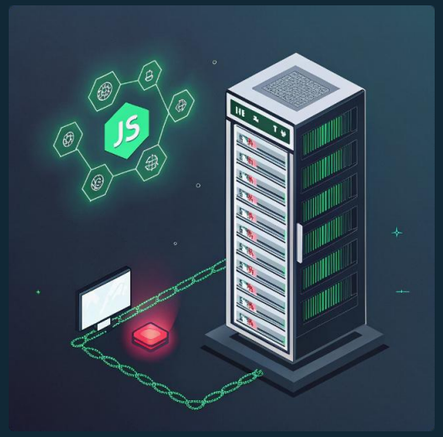
CollabCode ensures a smooth data flow from your browser to secure cloud storage. Every change is permanent and instantly accessible to your entire team.

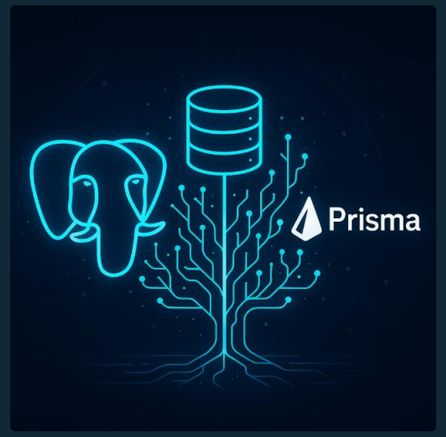
Key Principle: Forget local files. All your code lives securely in the cloud, guaranteeing ultimate consistency and reliability for every team member.

ARCHITECTURE

CollabCode's Technical Architecture







Frontend

- React + TypeScript
- Monaco Editor (VS Code)
- TailwindCSS

Backend

- Node.js + Express
- JWT Authentication

Database

- PostgreSQL
- Prisma ORM

> The Backend serves as the single source of truth, guaranteeing unparalleled data consistency and reliability.

Professional Editor Experience

CollabCode brings the power of VS Code directly to your browser, featuring Monaco Editor for an unparalleled, professional-grade editing experience.

Core Features

- **Advanced Syntax Highlighting:** Multi-language support.
- **Intelligent Code Assistance:** Formatting and IntelliSense.
- **Developer-Optimized Dark Theme:** For reduced eye strain.
- **Seamless Navigation:** Efficient multi-file workspace.

Smart Autosave

Visual indicators ("Saving" and "Saved") ensure your work is always protected, providing instant peace of mind and data integrity.

SAFETY FIRST

Ironclad Data Protection & Version Control

Comprehensive Protection

Every change is automatically tracked and stored centrally, providing a complete snapshot-based version history for easy recovery.

Why it matters: Eliminate the fear of irreversible mistakes. Teams gain confidence to innovate freely, knowing every version is recoverable.

01

Real-time Auto-Save

All changes securely stored in a central database.

02

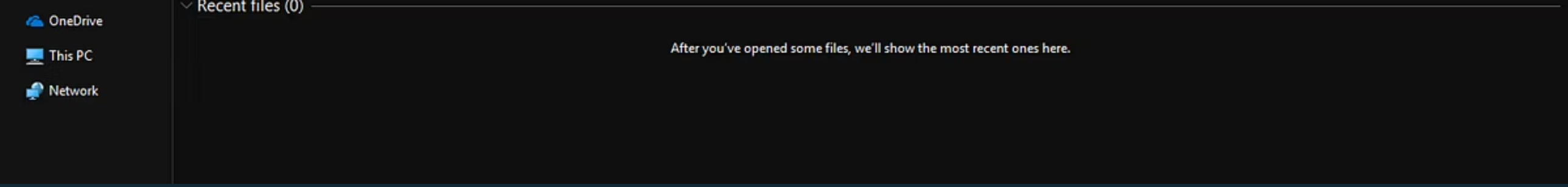
Comprehensive Version History

Snapshot-based history preserves every iteration.

03

Effortless Recovery

Preview and restore any version without data loss.



Intuitive Workspace for Seamless Collaboration

Experience a powerful, repository-style workspace designed for familiarity, professionalism, and enhanced productivity from the very first interaction.

Effortless File Navigation

Navigate your project with an intuitive file browser, featuring a clear visual hierarchy for easy access and organization.

Instant README Previews

Access instant markdown rendering for clear, up-to-date project documentation, ensuring everyone is on the same page.

Guided Onboarding

Onboard quickly with guided empty states that provide clear, actionable instructions, empowering users from day one.

Focused File Editing

Concentrate on individual tasks with file-level editing that opens only selected files, minimizing distractions.

Designed for immediate familiarity, our workspace minimizes onboarding friction and maximizes team productivity.

UNIQUE VALUE

CollabCode: Redefining Collaborative Development

Unleash Asynchronous Productivity

Collaborate on your schedule, eliminating real-time coordination bottlenecks and maximizing efficiency.

Uninterrupted Persistence

Every change is instantly saved and always accessible, guaranteeing no lost work and seamless continuity.

Instant Setup, Zero Local Dependencies

Dive into coding immediately with no machine setup required, freeing you from local environment complexities.

Integrated Version Control

Track changes, manage branches, and review code effortlessly, all within a unified and intuitive environment.

Our Core Advantage

Browser-based VS Code-grade editing, built upon an unwavering commitment to data integrity and team trust.

CollabCode eliminates the friction of traditional development collaboration. No more waiting for teammates to come online. No more lost work when someone disconnects. Just reliable, persistent, and professional code collaboration designed for the modern team.

Try CollabCod

View Documentatio

🔗 Project Resources

📂 GitHub Repository

Source Code & Documentation

👉 https://github.com/Nisarg9072-god/collab-code

🎥 YouTube Demo Video

Full Project Walkthrough

👉 https://youtu.be/LgaGGgIIbfY?si=LIq-RBmll0_ZkinZ

Thank you